

Name: _____ Reg #: _____ Section: _____

National University of Computer and Emerging Sciences, Lahore Campus

Course: Artificial Intelligence

Program: BS(Computer Science)

Duration: 45 Minutes

Paper Date: 7-May-2026

Semester: Spring 2026

Course Code: AI2002

Section: BCS-6G

Total Marks: 20

Page(s): 2

Exam: Quiz 3

Q1) A hospital AI predicts disease risk using patient health indicators.

(3+2+3=8)

Input Features:

x1: Blood pressure deviation index

x2: Cholesterol level index

For a patient:

$$x_1 = 1, x_2 = 2$$

Hidden layer:

h₁: w₁₁ = 0.5, w₂₁ = -0.3, b₁ = 0.1

h₂: w₁₂ = -0.4, w₂₂ = 0.6, b₂ = -0.2

Activation: tanh

Output layer:

o: w₁ = 0.7, w₂ = -0.5, b = 0.2

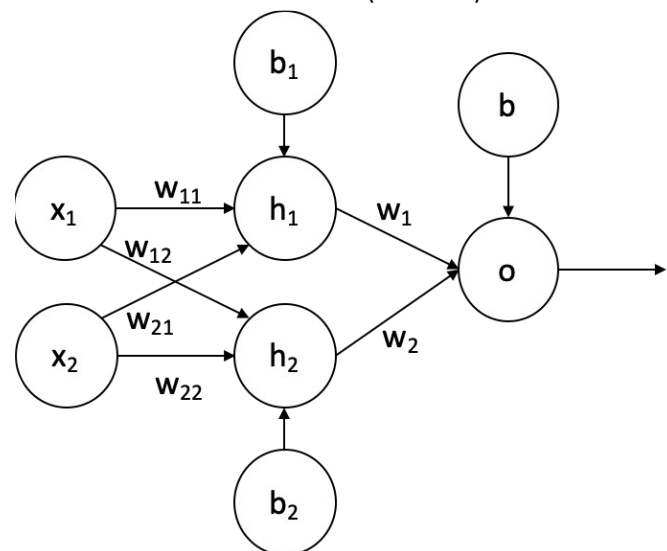
Activation: tanh

Target:

$$y = 0.8$$

Loss/Error: MSE

Hint (Derivatives): $\tanh'(z) = 1 - \tanh^2(z)$



(a) Forward pass (compute all neuron outputs)

Unit	Net Input	Output
h ₁	$(1)(0.5) + (2)(-0.3) + 0.1 = 0$	$\tanh(0) = 0$
h ₂	$(1)(-0.4) + (2)(0.6) - 0.2 = 0.6$	$\tanh(0.6) \approx 0.537$
o	$(0)(0.7) + (0.537)(-0.5) + 0.2 = -0.0685$	$\tanh(-0.0685) \approx -0.068$

(b) Compute gradients for output weights w₁, w₂

	Gradients $(o - y)(1 - o^2)$ (input)
$\partial E / \partial w_1$	$(-0.068 - 0.8)(1 - (-0.068)^2)(0) = 0$
$\partial E / \partial w_2$	$(-0.068 - 0.8)(1 - (-0.068)^2)(0.537) = -0.464$

(c) Update output weights and bias using learning rate $\eta = 0.1$

	New weights
w ₁	$0.7 - 0.1(0) = 0.7$
w ₂	$-0.5 - 0.1(-0.464) = -0.454$
b	$0.2 - 0.1(-0.864) = 0.286$

Q2) A text **clustering** system analyzes 5-word phrases using character-level **Hamming distance** (number of mismatched letters in the last word only; all earlier words are identical and ignored). (2+6 = 8)

The dataset is:

- A: MACHINE LEARNING WORKS FAST
- B: MACHINE LEARNING WORKS FAST
- C: MACHINE LEARNING WORKS WELL
- D: MACHINE LEARNING WORKS BEST
- E: MACHINE LEARNING WORKS WELL

Hint: (Hamming Distance between FAST and BEST is 2)

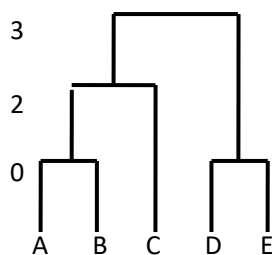
Clusters (A,B) and (C,E) are formulated at distance 0, due to no mismatches. The distance matrix is the same for single, average and complete link approaches at the intermediate iteration such that (A,B), D and (C,E) are the clusters.

(a) Fill in the distance matrix at this intermediate step.

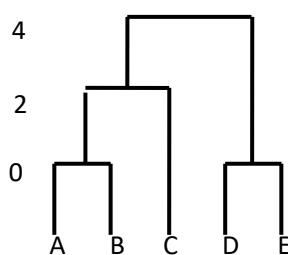
	A,B	C,E	D
A,B	0	4	2
C,E	4	0	3
D	2	3	0

(b) Based on the above distance matrix compute the remaining order of merging using single, complete and average link and label the respective dendrograms

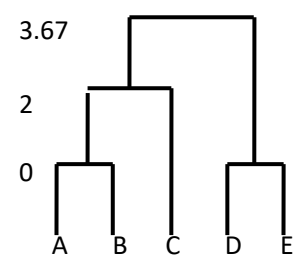
Single Link



Complete Link

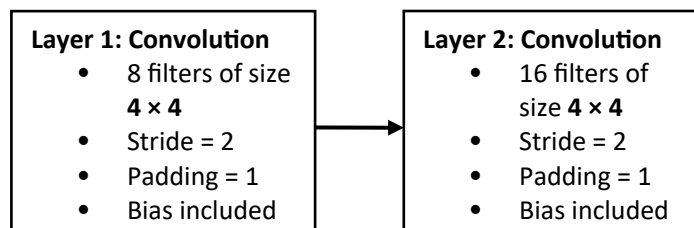


Average Link



Q3) An RGB image of size $64 \times 64 \times 3$ is passed through the following CNN:

(4)



Calculate the following:

	Layer 1	Layer 2
Output Dimensions	$32 \times 32 \times 8$	$16 \times 16 \times 16$
Parameters	392	2064